

Configuration-level split aliasing in object-detection benchmarks: what the reported split is selected on, and how much of the selection survives an independent split

Abstract

Object-detection papers report a number from a split they call held out. We audit that premise, in three results.

1. The convention is broken in our own corpora. 6 of 14 corpora give the reported number no independent held-out split: the checkpoint was selected there. Training and evaluation do not overlap — **checkpoint selection**, not contamination.

2. The direction varies by corpus. The premium is **positive in all four cells** (§5.1; the baseline arm gains more), the **paired Δ gap** changing sign by corpus (**not the selection term**); a declared, **descriptive** 13-cell family leaves **3** after BH; **five current-generation coders** agree at κ **0.824** and, re-verifying each row, **independently reach 19 of 30 corrected cells** (eleven unanimously), **26 of 76** stay undecidable; the printed **12–14 of 19** rest on **one rater**, **10–11** after re-verification (§6.1, S12). **The mean direction is metric-robust; one cell's significance and sign unanimity are not** (smoke2sf +0.063 pp, $p = 0.54$, 6+/5–).

3. Partial realization on a disjoint split. Separating the selection set from the reported set, the premium is realized only in part: the **claim-level interval includes zero** ([–17.3 %, 53.2 %], $P = 0.118$); on a difference scale **–1.367 pp** (CI [–1.59, –1.15]); run-level **[8.4 %, 37.4 %]** ($P = 0.002$, a bootstrap tail proportion) and leave-one-out **9.77 %–34.16 %**; the arm-cell and run-weighted point estimates are **descriptive**, given only with intervals (Table 4).

Replication lowers it 9 %–29 % (§8); no power analysis was registered, so the nine $n = 3$ cells stay descriptive.

Keywords: evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit; reproducibility

1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not chosen on that same split. Binding **val:** and **test:** to one directory breaks it silently — the pipeline

evaluates there, keeps the best-scoring checkpoint and reports its score: a **maximum**, not a sample. Surveys of the detection literature [1] trace how that number became the field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**, where we count how many of our corpora alias the two keys (19 YAML files, counting by file); **19 public benchmarks**, audited row by row for which split the reported number comes from; and **20 published third-party training logs**, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **measure the premium's fate** under a protocol separating the selection set from the reported set (**best** and **last** checkpoints and both splits exist, so the full 2×2 grid is computable): how much of the **val**-selected premium is realized on an independent split — the measurement this paper contributes and §3.1 formalises. The protocol is the companion's and is **cited, not restated**. And we **pre-register** a generalization hypothesis, then report that it misses its criterion — a registration **also reported in the companion**.

1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is defensible and they disagree by a factor of six: 2/19 (**structural**: one path bound to both keys), 10/19 (**'release' unit**: no independent held-out test), 13/19 (**'protocol' unit**: local independent testing gated), **12–14 of the 19** (**'reported' unit** — **12** the single rater's printed coding, **14** that rater's rows after a **non-blind recoding** moved two; **five blind coders agree at κ 0.824** and, once the record is re-verified first-hand, independently reach **19 of the 30** corrected cells, eleven unanimously (**S12**); the reported number's split is non-independent). These are defensible answers, not the four counting units (§6.1: 10/19, 13/19, 4/19, 12/19). Declaring the unit is therefore a methodological premise, not a caveat (§3, §6).

1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the training and held-out evaluation images is **zero** (SHWD excepted, where **train** and **val** are one directory). We do not claim a single bias

direction: our paired analysis finds **two corpora where `val` understates and two where it overstates**, and **nine cells are not testable at $n = 3$** (§5.1, §7.1). Nor do we claim the convention is framework-general (§6.5) or that the audited benchmarks are wrong: they report what their configuration computes.

Positioning (one sentence). We claim neither to be first to audit split problems in detection benchmarks nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level aliasing, a direction that varies by corpus, and the requirement that a count declare its unit**. Every number points to a local artifact; every claim narrowed after external checking is recorded as narrowed (§5.4).

Scope. All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**, the only family among the six frameworks we checked (five pinned repositories plus the Ultralytics release) selecting the checkpoint by the validation metric by default (§6.5) — stated at the claim, not only in a limitations section. **The registered multi-target replication is reported in the companion paper, and its first report, criterion and verdict stay there (§9):** this paper cites it for readability and **claims none of its results**.

2. Contributions and related work

Where this paper sits. Reporting practice has itself been audited: leakage and near-duplicate contamination [3], [4], [11]; split naming and the choice of standard splits [8], [9]; experiment- reporting standards [7]; checkpoint-selection criteria [6]; and budget-dependent rankings [2]. Each audits **data, reporting, or selection; none audits a release’s configuration binding** — the object of §3–§6.

2.1 The closest work, and where this paper sits (detailed)

- > The detailed comparison — the nine prior lines of work, what each measures, and the axis on which
- > this paper differs — is **Table S45 in the Supplementary material**; it is not printed here because
- > the journal counts text inside the 20–35 page limit and the comparison is background, not a claim.
- > **§2.2 (Contributions) and §2.3 (what we do not claim) are unaffected.**

2.2 Contributions

What this paper reports first (the boundary to the companion is in §2.3): the configuration-level audit of the `val/test` convention and its selection premium (§4–5); and the **measured** realization of that premium on a disjoint split with the reporting conventions it depends on (§8.6–8.7, §11, §3.1).

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium (best – last) is **positive in all four cells, the baseline arm gaining more (near-definitional; §5.1; four paired 95 % CIs excluding zero, §5.1 Tier 1) — consistent with the selection-bias literature [5], not new in kind. All four cells meet the stated-n criterion (n = 10–11)** and all four survive a Benjamini–Hochberg step over **these four** at $q = 0.05$ — the fourth, `dota15`, entered at $n = 3$ and was escalated to $n = 10$ under a criterion fixed in advance (**Table S28's note**) — so the convention systematically shifts **arm-to-arm comparisons**; its direction varies by corpus: three-way paired differencing gives 2 cells where `val` understates, 2 where it overstates and 9 not testable at $n = 3$ — the **no-systematic-direction** [10] reports, measured here on detection corpora.

We do not claim a general understatement (§5). It appears in third-party logs too (a searched 20-log sample; §5.5) — corroboration that the component exists elsewhere, **not independent evidence** and not a field-level rate.

Note: image-level leakage magnitude is quantified by the two 2026 studies; configuration-level aliasing magnitude has not been measured before.**

2. **We measure how far the selection premium transfers.** The measurement is the contribution; the protocol that makes it readable is the companion's and is **cited, not restated** here (§8.1). On the five arm-cells measured here the `val`-selected premium is realized on the disjoint split at

23.3 % with all five arm-cells at $n = 10$, and it is **not a constant**: the rate falls with the validation noise of the criterion used to select (§8.6–§8.7). The identity tying it to the winner's-curse share we **adopt** [13], not claim; the closest concurrent work [12, 15] reaches the same **direction** with a different quantity, and neither reports a **ratio of difference premiums** — $(T_{\text{best}} - T_{\text{final}})/(V_{\text{max}} - V_{\text{final}})$, on disjoint splits, in detection, on our own arm-cells, with measured `val` noise as moderator. A same-estimand ratio does exist elsewhere — in genomics, where it is written as a relative bias and

measured at 10–60 % (**Supplementary S12** carries the full citation). The replicate axis is **data order** (§8.4) — the **seed-budget discipline reproduced inside this setting** rather than a new one.

3. **We disclose our own defects, and the corrections that followed** rather than removing them: our analysis' first version had non-reproducible items — pod-dependent scripts, a citation to a file not holding the quoted numbers, no written endpoint convention. §10, not deleted.

2.3 What we explicitly do not claim (shared tooling)

The companion and this paper share project-internal process protocols — a multi-model review checklist and a local/cloud scripting discipline — and neither claims originality for them: **Supplementary S6, *Shared process protocols***.

3. Units: the methodological premise

"Have a split problem" is a property of a (**benchmark, counting unit**) pair, not of a benchmark. [14] **varies the definition on one file; we vary who controls the split and measure a different quantity: the share of `val`-selected premium that survives a disjoint split.** We use four units, defined by **who controls the split**:

- **`release`** — the split the benchmark's **own published release** ships: the `train:` / `val:` / `test:` entries and paths in the distributed archive. *Example:* an archive whose `data.yaml` binds `val:` and `test:` to the **same directory** records "yes" here (§4).

- **`protocol`** — the split the benchmark's **official protocol** tells you to use, which need not be the released one: a protocol may carve its own validation set out of the released training split.

Example: an aliasing release whose documented protocol still requires a held-out test split records "no" here and "yes" above — the units answer different questions.

- **`yolo\_dist`** — the split a **generic YOLO distribution package's** own configuration resolves to, i.e. what a practitioner taking the pipeline off the shelf gets. *Example:* a clean release whose community YAML points `val:` and `test:` at one directory records "yes" **only** here.

- **`reported`** — the split the number **the literature prints** came from (not the official leaderboard's own score), read from the release documentation and the

dataset's own paper. *Example*: two papers on one benchmark can differ here while agreeing under all three units above — hence the headline count uses this unit (§6.1).

A count is meaningless without its unit: the same nineteen benchmarks answer differently under each (Fig. 1).

The proposition is not ours to claim. That *the same data yield different defensible audit counts under different operational definitions* is stated independently for molecular benchmark auditing, in a 2026 preprint whose abstract closes "different defensible definitions return different answers from the same file" [14]. **The observation is the cited literature's; its instantiation in detection evaluation, and the four named units, are ours** — the units named by *who controls the split*, the 19-row evidence table, and the premium on **five arm-cells**. It is orthogonal to [15], which separates leakage by **type**; we separate the **counting subject**, so one type can still be counted under four units (§6.1).

> **Table S2** → Supplementary §S8.

The 'reported' unit is read from the release documentation and from the dataset's own paper: we did not sample papers per benchmark, so no prevalence claim is made (§11). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check **and one stated 'n_a' rule (Supplementary S12)** — the full 19×4 table and the cross-check against the original audit are **Supplementary S1**, the derived counts §6.1 and its table.

The rule itself is not new. The same toolchain's own course states it normatively — "*Test set is touched once, at the end. Not for tuning.*" and "*The splits must be independent on the dimension that matters at deployment.*" [16] — so what this paper adds is not the rule but the **audit and the count** under it.

The premise we rely on throughout: a "yes" under any unit means *the reported number is not a sample from an independent split*. **What that does and does not license is stated where the counts are printed** (§6.1 and the note under its table).

3.1 Formalisation: what the four quantities are, and what they can and cannot identify

Setup. Training is a randomised map from (seed ω , recipe θ , checkpoint e) to weights, and evaluation a fixed functional on a labelled split; write $V(e)$ and $T(e)$

for a checkpoint's `val` and `test` readings and $\hat{e}$ for the `val` argmax. Sections 5–9 measure four quantities: the selection premium, Δ_{gap} , κ , the realization rate (**notation and formal statements: Supplementary S11**).

Definition — the raw gap is not identifiable from the pair alone. $G = V(\hat{e}) - T(\hat{e}) = \delta + o$ mixes a split-difficulty term with a selection-optimism term; the decomposition, **and the reading that a single-model endpoint pair is unable to separate its two terms, are [17]'s**, while its **strong form** — that *any* δ' fits the observation equally well — is **stated here as a formalisation rather than attributed to that literature. The pairing assumption is what makes Δ_{gap} meaningful**, failing by construction under aliasing so that §9 cannot be read as a clean-protocol result; the identity holds value by value on all 13 scanned cells and both escalated cells at $n = 10$ (§5.6).

Adopted identity, not a claim of this paper. With $V(e) = v(e) + \varepsilon_e$ and $T(e) = v(e) + \tau + \eta_e$, $E[\text{premium_test}] = E[\text{premium_val}] - WC$, $WC := E[\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$ — the argmax is over a **candidate set fixed in advance** and e_f is a **pre-specified member of it** (what the budget-fixed `last` gives) — so the realization rate is $1 - WC / E[\text{premium_val}] \leq 1$, with equality only if selection is driven by signal alone [13] (**Supplementary S11** states which part of that the archive can and cannot settle) — the optimizer's curse in this paper's notation, the frame for §8.6–§8.7. **Two consequences are testable on the archive** — the per-run inequality, and monotonicity in the `val` noise (**Supplementary S11**). A third form, that the premium scales as $\sigma\sqrt{2 \ln E}$, was **tested and withdrawn** (**Supplementary S11**); contemporaneous work measures $\approx 5\times$ on the same $\sigma\sqrt{2 \ln K}$ bound [18].

Corollary (the premium is mostly shape, and the peak is the worst place to stand). $\text{premium_r} = \text{shape}(E) + v_r$ holds **exactly**, $\text{shape}(E)$ being the mean curve's peak-to-final gap, so shape is a functional of the mean curve alone — **not** an independent check on σ — and $E[v_r] \geq 0$. On our 40-run four-budget series **the monotone criterion is met on the clean re-run** while the $\sigma\sqrt{2 \ln E}$ form is **outside** its pre-declared band, so that scaling **stays withdrawn**; the pre-declared shape control passes on the mean it specifies but fails per arm, and we report both readings — the with-early-stopping series carrying the pre-declared verdict — so **we do not claim an 'E' effect**. The checkpoint rule is a **reporting rule** (the top-5 mean beats the argmax on the

same 40 runs), with **three limits**: an empirical capture rate, a `val`-side cost the `argmax` does not carry, and a two-split, five-epoch-grid scope.

What is ours, and what is not. Of the statements that follow, the raw gap's definition [17], the impossibility result for endpoint-pair estimators [19] and the family-dependence procedure [20] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**. Full statements and proofs are in **Supplementary S11**, and the detail displaced from here in **Supplementary S12**.

3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first

The propositions of §S11 are testable, so we ran them on rented GPUs with the **criterion fixed before the runs**; **Supplementary Table S37** gives, for each of the six experiments, the criterion it was judged against. Counts below are over **experiments**, not criteria.

The two negative outcomes and the redone design are stated as such in Supplementary Table S37 (G1 is void; **two of the five judged experiments are negative** — G1' and G2; the values behind every verdict are in S31–S33, the G1 family in S12; the reading below is stated more weakly in **Supplementary S11, section 13.3**).

We then ran the clean test of G3's epoch-budget series on **one machine with early stopping disabled**, all four budgets present: C1 is **met**, C2 falls **outside** its band, and C3 **passes** on the mean $\hat{e}/E$ it specifies but not per arm — see §3.1 and **Table S34b**; **the pre-declared verdict stays with the with-early-stopping series**. The judged criteria carry one message, **at the level of sign, not of functional form**: the selection bias moves with the amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file — **the form itself was withdrawn** (§3.1). Two consequences do not follow: §4's configuration-level fact stands, while the stronger reading — that aliasing *adds* a measurable amount on top — does not.

On the shared 5-epoch grid the `val` and `test` peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs: where they agree the **median Δ gap is lower** (mean **+0.13 pp**, median **−0.20 pp**, $n = 7$; **post-hoc**), where they differ it is **systematically not** (mean **+0.91 pp**, $n = 33$; permutation $p = 0.009$, **post-hoc**; the median reading is $p = 0.0011$). **This is a**

partial predictor, not an explanation of the sign; its limits — including the four of seven agreement runs whose gap is non-positive — and the per-run table and test are in **Supplementary S11**.

4. Finding 1: our own corpora

4.1 How common the aliasing is

In **our own packaged configurations** — the YAMLs we train from, a different object from a benchmark's own release — **6 of 14** give the reported number no independent held-out split: **AI-TOD, DOTA v1, DOTA v1.5, Mendeley face-mask, SFCHD** (five binding `val:` to `test:`) and **SHWD** (which binds `train` to `val`; **S1** row 13). Of the remaining 8, **4** keep the keys genuinely distinct and **4** have no `test:` key at all — the three groups account for all 14, and this is **a count over the configurations we hold, not a prevalence, nor the benchmarks' own releases** (audited separately, §6.1). Counting by file, **19 YAML files** alias the two keys, in five locations; on the archived manifest subset the ratio is **8 of 28**.

For the corpora whose `val` and `test` are the same directory this is not contamination: their image-level intersection with training is **exactly zero**, so the aliasing changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question. **SHWD is the exception and is not folded into that statement**: with `train` and `val` on one directory, its training and evaluation images are the same files. **It still counts among the six, for the reason the list is about: its `val` is not an independent held-out split either**.

4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [21]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [22]). §10's first recommendation: make "are `val` and `test` the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

5. Finding 2: the selection premium — magnitude, direction, and mechanism

Evidence. `eval_validity/selection_premium_local_20260916.txt`, reading **569 archived runs** from the local archive (the local version is the 2026-09-16 repair).

Definition. $\text{premium} = \text{best-epoch} - \text{final-epoch}$ mAP50-95 **within a single run**. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [23] (early stopping, data order) and [24] (benchmark variance across seeds). Being a maximum over a noisy curve, it also carries the upward bias of any selected extreme [13]: what those works average over as uncertainty, we measure as a reproducible component. **The two arms.** Every cell is a paired comparison in which the **baseline** arm follows the published convention and the **strategy** arm is the intervention under test — in the cells reported here, raising the peak learning rate from **0.001 to 0.005** with everything else held fixed (batch 32, imgsz 640, SGD; YOLOv12n with the shape-aware IoU). The two arms run the same code and the same data.

> **Table S3** → Supplementary §S8.

5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: "the convention understates the effect" is wrong, and external checking refuted it (§5.4, II). **Endpoint convention (frozen):** `best - last`

within a single run, both endpoints reported for every cell; §7.2–§7.3 quantify the other convention.

Tier 1: the selection premium is positive in all four cells and the baseline arm gains more, consistent with the selection-bias literature, not new in kind; **on the monitored split that positivity is near-definitional** (best is that curve's own argmax), and **the finding we report is the paired arm difference and its test side (§8.6)**. Pairing brings **no** variance benefit: the paired *t* of 5.74 and the unpaired Welch *t* [25] of **5.68** agree — at $p = 0$, **standard arithmetic rather than an empirical finding of ours** [24]. **Permutation checks** agree with the *t*-tests at each cell's attainable floor (**Supplementary S12**).

Meta-analyses of published results have long quantified this optimism: development-set early stopping is measured to *overestimate realistic model performance*, by as much as **18.0 %** accuracy [26]; and a large Kaggle meta-analysis finds *little evidence of substantial overfitting* [27] — complementary rather than contrary, since its mechanism is holdout **reuse** and ours needs none.

Metric sensitivity. On `metrics/mAP50(B)` the mean direction holds in all four cells, but `smoke2sf` loses both its significance (**+0.063 pp**, $p = 0.54$) and its sign unanimity (**6+/5-**, not **11+/0-**): **the mean direction is metric-robust, one cell's significance and sign unanimity are not** (full mAP50 columns and both mAP75 columns — the mAP75 `val` side is a **5-epoch grid**, so a rate computed from it is **upward-biased** — **Supplementary S12**).

Table 1. Tier 1 — each arm's **peak-final gap** and the **paired difference** between arms, by cell; the gap is a within-run maximum (§5.1), so the finding is the paired difference. `n` counts paired units (ten file-order replicates, one unseeded — **Supplementary S12**); both differences hold at $n = 10$. **A different run set from the transfer measurement in §8.6:** these are the 569-run local archive under the published (aliased) protocol, while that measurement's premiums come from the clean three-way runs — the two tables' premiums are different quantities, not two readings of one. Source: `verify_two_critiques_20260916.txt`; the `dota15` row's $n = 10$ escalation is in Table S28.

Cell	n (paired units)	peak- final gap, base	peak- final gap, strat	**paired diff**	sd	t	p	per m. *p* (floor 2/2 ⁿ)	95 % CI
---	---	---	---	---	---	---	---	---	---
smoke2sf (strong)	11	+0.79 9	+0.38 2	**+0.4 16**	0.2 40	5. 74	1.9× 10 ⁻⁴	0.00 098	[+0. 255, +0.5 78]
shwd2sf (corroborating)	11	+1.03 8	+0.51 2	**+0.5 26**	0.2 07	8. 41	7.6× 10 ⁻⁶	0.00 098	[+0. 386, +0.6 65]
a2d15 (directional probe)	10	+0.53 4	+0.26 2	**+0.2 72**	0.1 83	4. 69	1.1× 10 ⁻³	0.00 391	[+0. 141, +0.4 03]
dota15 (within-domain)	10	+0.94 1	+0.49 2	**+0.4 49**	0.1 96	7. 23	4.9× 10 ⁻⁵	0.00 195	[+0. 309, +0.5 89]

Table 2. Tier 2 — the paired difference Δgap by corpus. $\Delta\text{gap} > 0$ means `val` understates the strategy arm's effect. **Δgap is the split-gap arm difference**, not the selection effect: it decomposes into a difficulty gap and a selection term, and

Δ gap's sign is not the selection term's sign (Fig. S1). Equal split difficulty across arms is an **assumption we state**; the selection contribution is the **selection-action contrast** Δ prem_val - Δ prem_test. **'Reading' names Δ gap's sign, not the selection term's** (smoke2sf: Δ gap +0.461, selection -0.580). Sources: xeval_analysis_20260916.txt (the two val-understating cells) and x1_tier* / teval.csv (the two val-overstating cells).

Corpus	n	Δ gap (pp)	p	95 % CI	Reading (** Δ gap's sign**)
---	---	---	---	---	---
shwd2sf	10	**+0.779**	0.0053	[+0.297, +1.261]	`val` **understates**
smoke2sf	10	**+0.461**	0.0263	[+0.068, +0.855]	`val` **understates**
p_aitovis	10	**−1.226**	1.3×10^{-8}	[−1.370, −1.081]	`val` **overstates**
p_vistod15	10	**−3.315**	6.5×10^{-9}	[−3.676, −2.953]	`val` **overstates**

> **The split of Δ gap is recorded for two of the four cells:** difficulty +0.759 / selection +0.020 (shwd2sf), and difficulty +1.041 / selection -0.580 (smoke2sf) — in the second the selection term is **opposite in sign to Δ gap**, so the sign does not carry it, and **both intervals include zero** (Supplementary S9, Fig. S1). The two p_ cells' prem_test was **not recorded at that time**: their per-epoch test curves were evaluated on 2026-09-30, so nothing is imputed and the values are now recorded (Supplementary S12, Table S38).

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> **The four rows below are an escalated base, not a sample:** the two p_ cells reached n = 10 because their n = 3 Δ gap had been flagged, so the 2/2 balance is a property of the escalation rule; the other nine cells are **descriptive** (Supplementary S12). This cell is also the one where **3 runs** changed training path silently under GPU memory pressure, and why we do not call it clean is in **§10, defect 10**.

(Third §5 table, S5 → Supplementary §S8.)

Tier 2: on the same three-way corpora the paired difference Δ gap varies in sign. The raw $M_{\text{val}}(\hat{e}) - M_{\text{test}}(\hat{e})$ is **not identifiable**; the paired difference removes the difficulty term **under the equal-difficulty assumption** (Table 2) —

$\Delta_{\text{gap}} > 0 \Rightarrow$ 'val' understates. Its effective base is four cells, not thirteen, and an **escalated base** rather than a random sample, for the reason Table 2's note gives; "2 understate / 2 overstate" is therefore a statement about these four cells, not a rate for the family. No prospective power analysis was registered; the scan is in **Supplementary S12**.

Both escalations met the criteria fixed in advance. dota15 +0.449 pp (10+/0-); p_aitovis -1.226 pp and p_vistod15 -3.315 pp (0+/10-), permutation p = 0.001953, the attainable floor at n = 10. The best.pt endpoint was identified, not assumed: on test it reproduces the archived sio_b_results.csv to 0.0046 pp.

The formal claim: the convention systematically shifts **arm-to-arm comparisons** (Tier 1: four cells, CIs excluding zero); **the direction of that shift varies by corpus** (Tier 2: 2 understate / 2 overstate on the four-cell base) — **neither "a general understatement" nor "no systematic direction"**. BH at q = 0.05 **now leaves three cells significant** — p_vistod15, p_aitovis, shwd2sf [28], reversing what we reported before the escalation. The two families are not the same test: the 13-cell scan is descriptive, the confirmatory family is the four escalated cells (declared before their runs, §3.2) — the direction statement rests on those four.

Scope (cross-framework boundary). The convention this paper audits belongs to the **YOLO lineage** — Ultralytics YOLO [16] and YOLOX [29]: at pinned commits, Detectron2 (TEST.EVAL_PERIOD = 0) [30], MMDetection [31], DETR [32] and PaddleDetection [33] **report the final epoch instead** (§6.5) — that pipeline family's convention, **not the field's**.

5.2 Two versions must be disclosed side by side

Two archives, two versions of this table. The main text uses the **local archive** (569 runs, n = 11); the pod root (347 runs, n = 10) is a **Supplementary S12** cross-check agreeing in sign and magnitude, not in value.

5.3 The dota15 cell: now reproducible, and one wording defect exposed

This cell was not reproducible from the release as first written, and the repair exposed a wording defect (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

Supplementary S12 carries the two external critiques in full. **We used to write** that the gap was simply unidentifiable; the external check **corrects this to** the paired-difference form, which cancels the difficulty term under the pairing assumption stated in §S11. The argument and the recomputation are in **Supplementary S12**.

5.5 The same component in third-party public logs

The same quantity appears in other people's logs — a convenience sample with no sampling frame, not a field-level rate: 20 logs from 6 sources found by repository **search**, median **+1.04 pp**, max **+5.65 pp**, 18/20 positive. The queries, the rest of the frame and the reason public logs **cannot establish aliasing** are in **Supplementary S12**.

Five limitations: ① the two cells escalated in §5.1 reach $n = 10$ with **local** runs and test-side evaluations, and the remaining 9 cells' **test** side was closed on 2026-09-30 on one machine; ② **prem_test** is **best – last**, while the per-epoch test curve now exists for **all thirteen** corpora (**Supplementary S12**, Table S38); ③ **last.pt**'s val recomputation differs from the final **results.csv** row by ~ 0.02 pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — **prem_val** spans **+0.605 to +1.469 pp** across the five arm-cells (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points.

5.6 Why Δ gap's sign varies by corpus: three of five candidate mechanisms are refuted by the data

Three of five candidate mechanisms are refuted, and the mechanism itself is not identifiable — the sign of Δ gap is a sum of same-order terms, so no single mechanism can be read off it (§S11, Remark 3) — **but the paired quantity is, and is what we report**. The examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–S10.

6. Finding 3: how general this is across public benchmarks

Evidence: **benchmark_split_audit.md**, a line-by-line audit of 19 benchmark rows, plus **split_units_19rows_20260916.md** (recomputation script; no hard-coded percentages).

The nineteen audited benchmarks, each read from its own release or dataset paper: COCO [34], PASCAL VOC [21], Objects365 [35], Open Images v7 [36], DOTA v1.0 and v2.0 [37], VisDrone-DET [38], AI-TOD [39], UAVDT [40], xView [41], DIOR [42], NWPU VHR-10 [43], SHWD [44], SFCHD [45], MAFA [46], Mendeley face-mask [22], WIDER FACE [47], CrowdHuman [48] and D-Fire [49] (**per-unit markings**: Supplementary S1) — **the configurations this archive could read, not a sample of the field**. The question is **evaluation validity**: dataset-level bias auditing [50] and classifier-accuracy estimation [51] ask whether a reported number measures what it is taken to measure, but neither audits a release's configuration binding.

6.1 The result under four units (single-rater judgements)

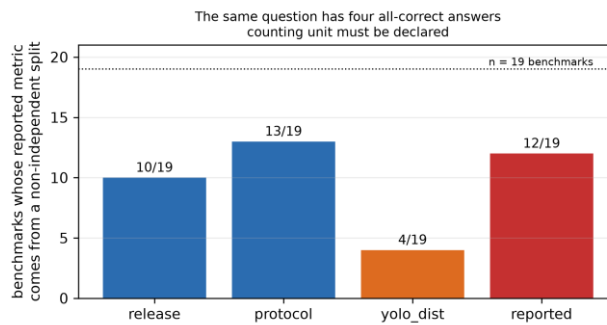


Table 3. The four counting units, with denominators named (§3). **Co-headline (2026-10-10):** the same four units were re-coded blind by **five current-generation coders** (κ 0.824; **Supplementary S12**), and re-verifying every row first-hand changed **30 cells** — so the table prints **both tallies**: the frozen single-rater counts, and beside them the same four units **re-tallied from the marking table after the 2026-10-09 first-party correction record**, under the **same per-column rule** (the value-to-count mapping is in **Supplementary S12**). Source: split_units_19rows_20260916.md. **Rows marked 'n_a' enter the denominator, not the numerator; alternative readings of the same rows are 12/19 for release, 12/15 the 19 less the four undecidable (n_a is two rows), and 13/19 for reported under the COCO-val reading. The 'clean' row is a reading: the audit counted 5/19, a later pass moved two, so this table reads 3/19. The 'reported' row's two denominators: 12–14 of the 19, and 12/15 = 80 % over the fifteen neither n_a nor undecidable (4 rows are evidence-insufficient). Both ranges span two readings, not intervals, on a single rater's judgement (per-coder κ 0.11–0.44, the weakest unit in the three-valued**

round). **Independent re-coding reproduces it only at slight-to-fair agreement, and ten coders given the archived evidence still do not reproduce the `reported` column (Supplementary S12). A model-assisted, non-blind second coding is reliability evidence, not a replacement (Supplementary S12).**

Unit	non-independent: printed (single rater)	non-independent: after first-party re-verification	of decidable rows: printed → re-verified	undecidable: printed → re-verified
---	---	---	---	---
release (the benchmark's own release)	**10/19 53 %**	≈ **14/19**	10/19 → **14/19**	→ —
protocol (the official protocol)	**13/19 68 %**	≈ **14/19**	13/19 → **14/19**	→ —
yolo_dist (the generic YOLO distribution package)	**4/19 21 %**	≈ **2/19**	4/15 ≈ 27 % → **2/17 ≈ 12 %**	4 → **2 rows**
reported (the layer actually reported)	**12–14 of the 19 ≈ 63–74 %**	**10–11 of the 19**	12/15 = 80 % → **10/16–11/17 ≈ 65 %**	4 → **2–3 rows**
↳ of which **literal two-key same path** (= the 2/19 the original audit reported)	**2/19 11 %**	≈ **1/19**	2/19 → **1/19**	→ —
↳ of which `reported` is **`clean`** (independent)	**3/19 16 %**	≈ **6/19**	3/15 → **6/16–6/17**	4 → **2–3 rows**

> **What a count here is, and what a "yes" means.** Every row is a **single rater's judgement**

> (§6.1), so the counts are printed as counts — the percentages beside them are a reading aid, not a

> rate estimated from independent coding. A "yes" under any

> unit means the reported number is **not a sample from an independent split**; it does **not** mean

> the benchmark is broken or the experiment dishonest, and it does not license a generalization claim.

These 19 rows were coded by one rater — 28 literal judgements (§10, defect 5) — and re-coded blind by **five current-generation language models**, who agree at **Fleiss κ 0.824**; re-verifying every row first-hand changes **30 cells**, and the coders **independently reach 19 of 30 (63.3 %)**, eleven unanimously, while **26 of 76 cells stay undecidable (S12)**; a **non-blind, model-assisted** pass agrees at κ **0.874 (Supplementary S12)** and **moves two rows**. reported means **the number the literature prints**: it is printed as a range, **12–14 of the 19 $\approx$ 63–74 %**, with the **80 %** column the lower end over the **12/15** decidable rows, and **n_a** rows counted in the **19** but not the numerator (**Supplementary S12**).

Five clauses move to Supplementary S12 (the self-check's column and subtotal equalities to

S1): the COCO row's judged number, the `yolo_dist` first-version correction (**4/19**), the literal-two-key self-check (**2/19**), Mendeley face-mask's **generator** provenance, and the audit's unused "63–79 %" bound.

6.2 "Same directory" has four levels, and the audit reported only the narrowest

The levels, the rows each admits and the resulting shift are in **Supplementary S12**, Table S12.

6.3 We do not reproduce the original audit's two numbers, and we say why one rests on the narrowest reading of "same directory", the other on a stale snapshot; both recomputations are in **Supplementary S12**, Table S13.

6.4 One judgment the evidence does not support (corrected)

One classification correction is recorded in Supplementary S12 (D-Fire moves to the clean group).

6.5 Across frameworks: is the convention specific to YOLO?

Among the pinned default configurations we checked, only the YOLO lineage does this by default. At pinned commits Detectron2 (TEST.EVAL_PERIOD = 0), MMDetection, DETR and PaddleDetection **report the final epoch instead** (file-by-file evidence, 25 configuration files: **S12**, Table S14; **a targeted sample at pinned commits, not a census**, so "the convention" means that family's, not the field's). A **cross-lineage realization test** — a Faster R-CNN (ResNet-50 FPN) detector, a family that **reports the**

final epoch by default — on VisDrone under the same three-way split, at **six seeds**, is **positive in all six runs**, with a **95 % interval excluding zero** (68.2 % realized; +0.0809 pp mean prem_val, 95 % CI [26.9 %, 123.5 %]) (values, interval and per-run table: **Supplementary S34**): the mechanism is **not confined to the YOLO lineage in this instance**. A **second detector family** (RetinaNet, both corpora) keeps the **direction** but not the strength (**Supplementary S34e**). A second corpus, same protocol, same direction: **6/6** and **12/12** runs. This is **one detector, one protocol, two corpora, one setting** — **not a generalisation beyond the YOLO lineage** (**Supplementary S34c**).

7. Finding 4: what "significant" means, and how the endpoint convention changes it

7.1 With $n = 3$ a permutation test cannot reject in principle

A power note for the cells that remain at $n = 3$. Recovering each Tier-2 cell's sd from its own printed 95 % CI gives **0.20–0.67 pp**; at $n = 3$ the required effect is **0.65–2.19 pp** at $\alpha = 0.05$, two-sided — **at or above every Tier-1 effect here** (+0.27 to +0.53 pp) — and 0.2–0.7 pp at $n = 10$. The nine cells at $n = 3$ are therefore **not testable at this design**, and are reported descriptively only. A **prefix- n table ($n = 3 \dots 10$)** and the derivation with the non-central- t factor ($3.26 \cdot \text{sd}$) are in **S12**; the permutation floor $2/2^n$ binds at $n = 3$, not the t test.

For a sign-flip permutation test on n pairs the smallest attainable two-sided p is $2/2^n$: at $n = 3$ that is **0.25**, so such a cell has **no resolution at this n** ; after §5.1's escalations **no headline cell (Tables 1–2) is $n = 3$** (the derivation and each n 's floor: **Supplementary S12**).

7.2 The two clean-protocol cells clear significance by $n = 6$ –8

The weaker endpoint reaches $*t^*$ -significance at $n = 6$ on ``shwd2sf`` ($p = 0.0164$) and $n = 7$ on ``smoke2sf`` ($p = 0.0400$); the exact permutation — **floored at $2/2^n$ (0.25 at $n = 3$)** — follows one step behind on `shwd2sf` and none on `smoke2sf` ($n = 8$ on both ``selected`` readings; **Table S42** gives every prefix n , both endpoints). The clean-protocol seed extension that reaches $n = 10$, and the per-cell numbers, are in **Supplementary S12**** and **Table S15**.

> **First-report attribution (a convention we hold to).** Three things here belong to the companion

> paper and are cited, not claimed: the two Δ values (**+1.434 / +0.507 pp**), the three-way remediation

> protocol itself, and the pre-registered multi-target replication (§9). What this paper reports first is

> ****the selection premium (§5), the 2×2 realization of that premium on a disjoint split (§8.6–8.7), and**

> the formalisation of the realization rate (§3.1)**. The full statement — including how the companion is

> named to the editor while it carries no identifier — is in **Supplementary S12**.

7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

One run is recorded with two different endpoints in two archives — which checkpoint is recorded was never specified: **Supplementary S12**, Table S16. Taking the final-epoch entry (67.12) gives $\Delta = -1.490$, $p = 0.0141$. The best-epoch entry (70.06) reads $\Delta = -0.902$, $p = 0.2931$, **but had been computed with the run's 'last' reading**; corrected 2026-10-02 it becomes $\Delta = -1.490$, $p = 0.0141$, as released. So under the **'best'** convention of §5 this cell is **significant**, and nothing printed here moves.

8. Finding 5: how far the selection premium transfers to an independent split

8.1 The protocol is the companion's; the measurement under it is this paper's

The three-way remediation protocol is **defined and reported in the companion paper** — which first reports that **the effect reproduces in direction while its magnitude falls** — so this section **cites it** rather than presenting the replication. **In this paper's own terms it is three things**: a **val** carved from the pool so that training never sees it; the target corpus's own held-out **test**, with a filename-level check that its images do not intersect training; and an endpoint **defined in writing before any run**. The registration text and its **FROZEN-HASH** stamp are in §9 and in the author-side registration record; **Tables S24–S27** carry the registered cells' readings. What this paper measures **under** that protocol is the quantity §3.1 formalises: **how much of the 'val'-selected premium survives on a split no selection rule can touch**, and that measurement is this paper's.

8.2 The protocol's results on the companion's corpora: transferred

The protocol's results on the companion's corpora are the companion's first report. What this paper adds under the protocol is the measurement of §8.6–§8.7; the transfer (2026-09-21) is recorded in the module-transfer note of that date.

8.3 A direct measurement of data-order replication (not seed variance, and not the selection claim itself)

A direct measurement of data-order replication — reproduced inside this setting: adding replicates that permute the training file order lowers the mean by **9 %–29 %**, each figure taken **against that cell's own $n = 6$ mean** and not against the pooled headline effect, the **seed-budget discipline** measured rather than assumed ([52]'s setting, §8.4). **What these replicates are.** The per-run randomness is a **deterministic permutation of the training file order** (`--shuffle-seed`, the number in the run's name), **not** the framework's seed argument (Supplementary S12, section 9.3, with the two corpora whose replays are identical across `--seed` values). The quantity is therefore **data-order replication**, not seed variance: the 9 %–29 % bounds the **file-order** component of the reported mean, **not** the variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** (§8.6 positions the closest concurrent work, a dose–response over 2,047 datasets), so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards** (design and recomputation check: Supplementary S12, Table S18). **The title is deliberately the weaker, sample-scoped claim:** with three initializations the interval is wide, and extending to five does not narrow it (the gain SD reaches the data-order component's order), so the components are **consistent with, not established by**, these initializations.

8.4 The three variance components: the excluded one is the smaller in this sample — a sample-scoped ordering, not an established one

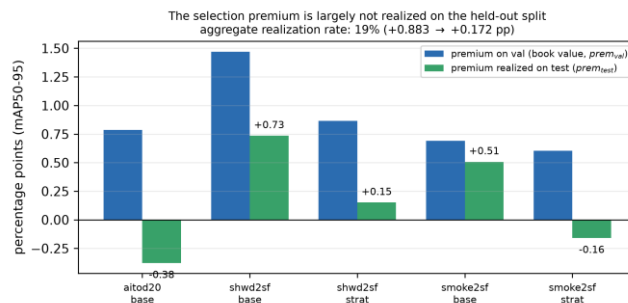
The three variance components, and the excluded one is the smaller in this sample: the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order — the **same ranking** as the shuffle-order result of [52] (Supplementary S12), so we **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is **sample-scoped** — at five initializations the gain SD moves **0.146 → 0.439 pp**

and its interval **overlaps** the data-order magnitude, so the two are of the same order rather than one clearly smaller — is in **Supplementary S12** and Table S19.

8.5 Why "same-pipeline pairing" is a necessity, not a habit

Pairing is not neutrality: the paired design covers the **budget and data** axes only, and **not** the optimiser's (why, and the counterexample: **Supplementary S12**).

8.6 The premium's realization rate: the claim-level interval includes zero ($[-17.3\%, 53.2\%]$, $P = 0.118$); the point estimates below are descriptive (arm-cell median 17.6 %, five arm-cells, $n = 10$; run-weighted 23.3 %)



What this section claims, and what it does not. The rate below is the realization of the val-selected premium on a disjoint split **for the five arm-cells measured here, under the companion's protocol — not a constant**. The escalated cells reach a **pooled 55 %**, but the two batches are **not pooled** (different runs, different protocol, §8.1). "The convention realizes about a quarter of the val premium" would be false; "in these five cells, about a quarter survives" is what we measured.

How much of the gain selected on `val` survives on `test`?

Table 4. The premium selected on val versus what that same selection action realizes on the disjoint test split, per arm-cell (**an unbalanced pool** — two corpora contribute both arms, aitod20 only its baseline — so the arm-cell is the unit). Source: the **per-epoch `val` curves** (results.csv, for prem_val), xeval_20260916/matrix.csv, and this revision's two batches (x4_teval.csv on A, x4fill_20260925.csv on B, for prem_test). **The §10 memory-pressure bound (defect 10) does not arise here:** its hits are in §3.1's clean batch and §5.1's p_vistod15.

Cell	Arm	n	`prem_val`	**`prem_test`*	**realizatio
------	-----	---	------------	----------------	--------------

---	---	---	---	*	n rate**
shwd2sf	baseline	10	+1.469	**+0.736**	**50 %**
shwd2sf	strategy	10	+0.865	**+0.152**	**18 %**
smoke2sf	baseline	10	+0.691	**+0.509**	**74 %**
smoke2sf	strategy	10	+0.605	**−0.157**	**−26 %**
aitod20	baseline	**10*	**+0.766*	**−0.218**	**−28 %**
		*	*		
aitod20	**strategy*	**10*	**+1.111*	**−0.543**	**−49 %**
	*	*	*		

> **The inclusion rule, stated as a rule.** An arm-cell enters the pool iff its `test` side reached $n \geq 5$ (work/item2_final_v2_20260925.py), applied to every cell alike; **exactly five met it at the 2026-09-25 freeze** (50 runs). The aitod20 strategy row **qualifies under the same rule but completed after the freeze**, so it is printed here as the sixth row and kept out of the primary pool — **both readings are reported and both intervals cover zero** (Table S20's note, §8.6b).

where $\text{prem_val} = V_{\text{max}} - V_{\text{final}}$ (selecting on val) and $\text{prem_test} = T_{\text{best}} - T_{\text{final}}$ (realized on test).

The three bases: run-level [8.4 %, 37.4 %], $P = 0.002$; five-cell [−17.3 %, 53.2 %], $P = 0.118$ (claim-level, **zero inside**); eleven-cluster [19.3 %, 60.4 %], $P < 0.001$ (a bootstrap tail proportion **at the 20,000-resample resolution**, i.e. below the floor this procedure can print).

Aggregate (runs weighted equally, $n = 50$): $\text{prem_val} +0.879$ pp $\rightarrow$ $\text{prem_test} +0.205$ pp. **At the claim level this is not significant:** the **claim-level interval** is the cell-level, five-cluster one, [−17.3 %, 53.2 %], $P(\text{rate} \leq 0) = 0.118$, zero inside. Descriptively, arm-cell median **17.6 %** and run-weighted **23.3 %** (span 2/5 negative: Table 4). On a difference scale the eleven clusters give **−1.367 pp**, 95 % CI [−1.59, −1.15] (five-cell denominators **+0.605 to +1.469 pp**). Re-aggregated: run-weighted **36.9 %**, arm-cell-equal **39.3 %**, 95 % CI [19.3 %, 60.4 %] — **completing the cell, not the weighting, is what moves it**: at $n = 10$ the three weightings agree, so **the fragile axis is pool composition**. **Leave-one-out 9.77 % to 34.16 %**. In two arm-cells the selection action **costs** rather than fails to pay off (smoke2sf strategy, aitod20 baseline); the escalated cells of §5.1 are **not pooled with these five** (different runs, different

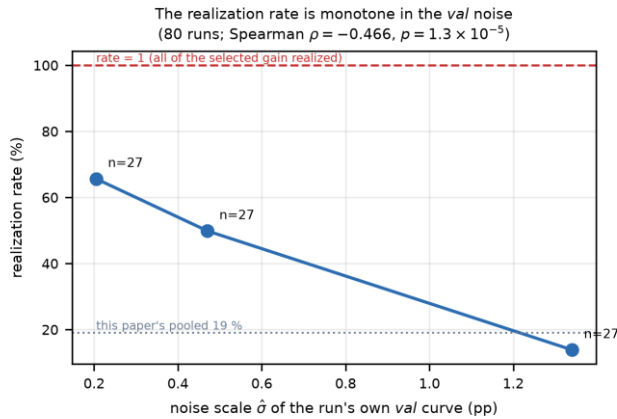
protocol). The closest concurrent work [12], the winner's-curse check, the withdrawn $\sigma\sqrt{(2 \ln E)}$ form and each cell's corpus pair: **Supplementary S12**.

(the **best.pt**-versus-argmax residual, our withdrawn attribution and why the remainder is **unidentified**: **Supplementary S12**.)

The monotonicity is a correlation, not a causal claim, and at the arm-cell unit the nine-cluster reading is $\rho = -0.433$ ($p = 0.244$) — see **Fig. 3** and **Supplementary S12**: the $n = 80$ are the 81 runs with a measured **test** side less the one whose **prem_val** is 0. It has a named precedent: [53] gives validation overfitting (VO) its name and bounds the expected best observed loss by $E[\min L] \geq v^* - s_m\sqrt{(2 \log(Nq))}$, an optimism growing with the number of selection queries Nq .

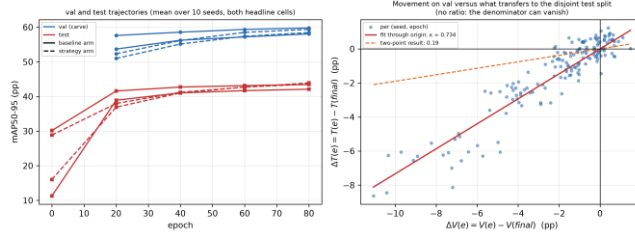
Recomputation status. The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; §10, **defect 5**); a GPU 2×2 **evaluation** on **41** three-way runs of **YOLOv12n** [54] $\times$ {**best**, **last**} $\times$ {**val**, **test**} makes them so, **retraining not**, reproducing the archived **best** $\times$ **test** values to within **0.0047 pp** (median **0.0021 pp**) — the endpoint rule's positive control (**Supplementary S12**).

Three limitations: ① in the **41-run epoch sweep** only the two headline cells reach $n = 10$; **aitod20** contributes one run of 30 epochs and 6 checkpoints where others contribute **20** (hence 806, not 820), its Table 4 arm-cell a separate $n = 10$ batch; ② **closed: every retained checkpoint is now evaluated**, the *stability* reading is narrower than the grid; ③ κ is relative to each run's own final epoch, not causal. The 806-row evaluation is archived (xeval_perepoch_20260918/, with a SHA-256 list).



Pool composition — and why the claim does not turn on it. Table 4 prints six measured arm-cells; adding the sixth (**aitod20 strategy**) moves the run-weighted rate **23.3 %** $\rightarrow$ **8.7 %** and the median **+17.6 %** $\rightarrow$ **-4.2 %**. **The inclusion rule is a rule, not a per-cell choice:** an arm-cell enters iff its ``test` side reached  $n \geq 5$` , and exactly five met it at the freeze, where all three weightings coincide (**23.26 %**). The sixth row **qualifies but completed later**, so it is the sensitivity. **Every claim is interval-level and any scope gives the same verdict** — five-cell **[-17.3 %, 53.2 %]** ($P = 0.118$), six-cell **[-4.9 %, +21.3 %]**, eleven-cluster **[19.3 %, 60.4 %]** — the cell-level intervals **both cover zero**, so **no conclusion depends on the pool boundary**.

8.7 The per-epoch view: the realization rate is not a constant



One metric self-correction first. Our first version used $r(e) = \Delta T(e)/\Delta V(e)$, producing values such as **-860 %** and **+1031 %**: at late epochs $V(e) \approx V(\text{final})$, the **denominator vanishing** — **an ill-conditioned metric, not a finding**. We report only the non-dividing $\Delta V(e) = V(e) - V(\text{final})$ and $\Delta T(e) = T(e) - T(\text{final})$, summarized by the through-origin slope κ of ΔT on ΔV .

> **Tables S21–S23** $\rightarrow$ Supplementary §S8.

Slope κ (how much of the movement on **val** transfers to **test**). On the **complete curve** —

$n = 765$ (seed, epoch) points over the 41 runs — **$\kappa = 0.708$, $R^2 0.937$** , with a run-clustered (sandwich) 95 % CI **[0.671, 0.745]**; the within-run estimator gives **0.698**. **Two controls (Table S43):** a per-run κ distribution, and a free-intercept fit whose **intercept is indistinct from zero** — so the through-origin constraint is **supported, not assumed**. Across the curve ΔV reaches **-8.1 pp** and about **71 %** transfers: **early-training progress is real progress**. §8.6's realized share measures the **selection-relevant sliver** near the peak, where ΔV is of order ± 0.2 pp — so

"most of the movement on `val` transfers" and "the little extra gained by selecting near the peak does not" are both true, and the premium is the residual of the transfer rate. And κ is not constant across arms either — in the direction the selection account predicts: baseline $\kappa = 0.600$ ([0.581, 0.619]) against strategy $\kappa = 0.767$ ([0.735, 0.800]), difference +0.168, 95 % CI [+0.130, +0.203].

A second result, and a self-correction the curve forces: the sign-of- Δ gap claim we wrote from four sampled epochs is too strong for one of the two cells (on the complete curve smoke2sf starts negative) and is narrowed here; per-checkpoint values: Supplementary S11.

The limitations of this section are in Supplementary S12.

9. The registered multi-target replication: reported in the companion paper

First report, criterion and verdict stay with the companion paper; nothing here is a claim of this paper. To read its registered table: ① it is a three-way split (train / selection / report); ② it ran under the protocol then in force, where the `val` side aliases the `test` half — the one fact §7 takes; ③ its criterion was frozen before any run existed (FROZEN-HASH 6a7eee7b3e34b15ce5adcba14cf7ea36) and its verdict entered is "criterion not met", deliberately not "hypothesis refuted" — the companion reads the outcome as "direction replicates, size does not". Record: Supplementary S24–S27 (source of truth: the companion's archive).

10. Our own defects (disclosed, not hidden)

#	Defect	Handling
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1	The clean-protocol results document **cites a ledger file that does not exist in the sense claimed ** : neither `_r10_final_stats.txt` nor the "real log" `_r10_stats_0403.txt` holds **any** headline value (6 probes each, **0/6 hits**) ⇒ **no surviving stdout ledger**	**We do not point at a replacement file** (renaming would manufacture a new false citation); the text says **"no stdout ledger"**, and the p-values are recomputable from `pvalue_table_local.py`. Detail: §S12
2	~the dota15 cell of `selection_premium.py` was not reproducible~ → **fixed** (stale root directory; the matcher assumed a seed suffix the run	Stated in §5.3; the "3 seeds" note was **retracted on 2026-09-21** — the third run's seed is s42, the

	names do not carry); it now **reproduces M3's +1.12/+0.95/+0.93 value by value**	campaign's fixed shuffle seed. Detail: §S12
3	**`sio_b_results.csv` has no written endpoint convention**: `t2_mask2mende_base100_s42n` is recorded as final (67.12) in one archive and best (70.06) in another	Stated in §7.3, with the consequence of each reading (significant versus not)
4	The two headline scripts **originally depended on a rented pod** (`import paramiko` plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release	**Fixed**: both now **read the local archive**; the main text uses the local versions and lists the pod versions as cross- checks
5	**One script now recomputes the σ decomposition** — the data-order and initialization components from the archived per-run values, each asserted against the printed number — but it can only **cite** the augmentation component, whose record holds an SD and no per- run values; **no frozen effect- size rule file**; **no script generates `benchmark_split_audit.md`** (hand-written, 28 judgments as literals, though its counts are recomputed)	The first is closed by script, with its one cited (not recomputed) component named; the other two stay labelled "manual / not archived"
6	The local archive **has no copy of 6 of the 7 corpora**, and of the three-way curve it holds **only the manifest** (`split_defs_20260916/`), not the other curve directories	**Partially repaired (2026-09-16, GPU run)**: the 41 runs' per- epoch curves, `sio_b_results.csv` and the ** 2×2 evaluation matrix** (164 evaluations) are now local; **retraining remains non- recomputable**
7	The audit document has **internal inconsistencies** (nine items, found by our own census)	Each adjudicated: **3 misreadings by the inventory, 4 real defects, 1 convention risk, 1 classification correction (§6.4)**
8	Our own corpora contain `val == test`	**That is this paper's argument**, not an

9 **Our first per-epoch report
 claimed curve-wide stability
 from a four-point sample**;
 the complete curve overturned it for
 one of the two cells (§8.7)

10 **Memory pressure silently
 changes the training path**:
 under GPU memory pressure
 Ultralytics switches
 implementation **without
 changing the recorded
 configuration** — a batch
 reduction, or a fallback of label
 assignment to CPU that can fork
 the trajectory — and case (a) is
 **invisible in the standard
 artifacts**.

embarrassment to be
 hidden
 **Corrected in place,
 with the retracted
 sentence quoted**:
 the claim is **narrowed**,
 not deleted, and the
 grid's apparent
 uniformity is named as
 partly an artefact of
 where it sampled.
 Detail: §S12
 **Disclosed, not
 retracted; we do not call
 the affected batches
 “clean”.**
 Configuration-level
 hits: **2** places,
 4 runs. **A
 reproducibility residual,
 not an event size**:
 same-config repeats are
 bit-identical
 (**0.000 pp**, four
 pairs, same machine
 and day), while the
 quantity a bound would
 be about — archived
 run against clean re-run
 — is **0.000 / 0.020 /
 0.150 / 0.000 pp**
 (**Table S41**), so
 ** ≤ 0.05 pp** holds
 only inside the clean E
 = 50 batch and is **not
 extrapolated**;
 without a no-event control
 the event effect is
 unidentified (such
 a control is itself **not
 reproducible across
 dates**, **0.440 pp**),
 and elsewhere it sits
 inside the zero-event-
 control spread. The one
 clear case is $\Delta V_{\max}$
 +0.769 pp. Event
 counts, per-run
 consequences and the
 log fingerprints are in
 **Supplementary S12
 (defect 10)**.

11. Recommendations

Four checks before reporting, each with the artifact that discharges it. ①

Are `val` and `test` the same source? Make it a **hard pipeline check** on every dataset YAML (§4.2); the output is the resolved paths. ② If you must select on the reported split, say so and report a selection-free endpoint — **the 'final'-epoch endpoint, the only selection-free one this archive measures** — alongside — the step **enlarges** the arm difference (§5.1). ③ State the test convention: below $n = 6$ a permutation test has no resolution; print the $2/2^n$ floor beside every p . ④ Declare the counting unit behind every count, as Table 3's header does.

⑤ **A plateau is a post-hoc candidate, not a rule we validate.** Weight averaging is established [55]; what is *measured here* takes the **mean of the ranked top-5 'val' epochs' readings** (a reading average, not a weight average), capturing **70 %** of the realizable gain against the argmax's **30 %**, at the cost of **0.73 pp** below the peak — a post-hoc reading of our own 40 runs, untested as an intervention. The per-rule table and the candidate mechanism are in **Supplementary S11**, Measurement 3.

12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.
- For several benchmarks the "generic YOLO copy" is **per-project**, not canonical.
- Under the reported unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do not guess. Counts are **row-equal over the 19**, and the grading enters the **error bar, not the count (Table 3's second column)**.
- Two evidentiary differences (pod versus local) are listed side by side.
- The census is a **literature/release survey**, weaker than our **self-measurement** (§3).
- The **difficulty/selection decomposition** (§5.6) covers **2 of the 13 cells** (the headline two); the **identity** behind it is checked on **all 13** (`work/gap_mechanism_20260916.txt`) — what is scoped is the decomposition, not the identity.
- The intervention axis the headline comparisons use is **'lr0' (0.001 → 0.005)**; two further orthogonal axes were run, one reproducing, as on `lr0`, the **corpus-**

dependent character of the shift (Supplementary S8, Table S40b); the cross-lineage probe is

one detector, two corpora, and incomparable magnitudes.

- **Four disclosed uncertainties** bound our claims: the 19-row audit is **single-rater**; the log sample is a **convenience sample**; the GPU memory-pressure effect is **bounded, not estimated** (≤ 0.05 pp where same-config pairs can be compared, fresh replicates being bit-identical; cross-date comparisons carry a **0.440 pp** drift); and the §8.3 replicates vary the **training file order**, not independent seeds. **And, not as an uncertainty: no interval here folds into §8.6's rate. Supplementary S12.**

13. Conclusions

A count is meaningless until its **unit** is named: the same nineteen benchmarks answer differently under each declared unit (Fig. 1, Table 3) — **10/19** under release, **13/19** under protocol, **4/19** under yolo_dist, and **12–14 of the 19** under reported (whose COCO-val reading is **13/19**). **Re-tallied after the 2026-10-09 first-party correction record, under the same per-column rule and with these printed counts frozen, the same four units read 14/19, 14/19, 2/19 and 10–11 of the 19** (Table 3, Supplementary S12) — the two **layer** units rise, the two **package and report** units fall, and the **majority** conclusion is unchanged (the weakest recomputed unit is still **10/19**). Three findings stand. **The most robust number is the transfer slope κ (§8.7): early `val` movement does transfer to `test`, and the premium is its residual.** The premium is positive in all four cells, the baseline gaining more, and the paired Δ gap **varies in sign** by corpus — so "the convention understates the effect" is not claimed. On five arm-cells at $n = 10$ it is realized on a disjoint split at **23.3 %**, interval containing zero. Two things do **not** hold: the $\sigma\sqrt{(2 \ln E)}$ scaling was tested and **withdrawn**, and no power analysis was registered, so the nine $n = 3$ cells stay descriptive. What we ask of a detection paper is narrow: **name the counting unit, freeze in writing which checkpoint is reported, and state the interval at the claim's level.** First report is the companion's (§7.2).

Declarations

- **Data availability.** The **seven experiment corpora are open**; what the pack lacks is their **images**, so retraining is not recomputable (§10 defect 6). Separately, §5.5's **third-party logs are not redistributed** (hashes + pointers

only), as **5 of their 6 sources state no licence**. The package — data, per-epoch matrix, verdicts, checkers, `MANIFEST_sha256.csv` — is at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29). The reproduction package itself — data files, per-epoch matrix, verdict files, checkers, with their SHA-256 list in `MANIFEST_sha256.csv` — is deposited and openly available at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29).

- **Recomputability boundary**. Every reported number is recomputable from the released per-run artifacts **without retraining**, each with its script; **three are not, and they are named**: the corpora and carve directories the archive **holds no copy of** (open corpora, images not in the pack; §10 defect 6), the conflicting-endpoint archives (§7.3), and the `aitod20` completion to $n = 10$.

- **Competing interests**. None declared.

- **One quantity, two spellings**. The $n = 10$ attainable floor is **exactly** $2/2^{10} = 0.001953$; the tables use the exact value, the abstract writes **0.002**. Same floor, spelling only.

- **Reproducibility**. Every number points to a local artifact (**artifact paths in this article and in the Supplementary are relative to the released `02\_release\_data/` directory**); **289 audit anchors** mechanically guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`, `verify_pulled.py`, `gap_mechanism_20260916.py`, `verify_pr_forms_20260917.py`, `verify_submission_pack_20260917.py`) ship with the archive: **one runs unaided for a holder of the archive; four are author-side gates; and the remaining one -- the anchor audit itself -- is partly runnable, checking 270 of the 289 anchors from the released copy alone (the other 19 need the authors working-tree inputs)**. The archive is the one deposited above.

References

> **Note on preprint citations**. Seventeen of the 55 entries give an arXiv edition; **three** also name a peer-reviewed venue and **fourteen** have none. Those fourteen cover four benchmark or dataset canonical releases, two methodological works and eight concurrent or tooling works; **all are cited for positioning only**, and every number comes from our own corpora or the third-party logs of §5.5. The full enumeration is in the supplement.

- [1] Z. Zou, K. Chen, Z. Shi, Y. Guo, J. Ye. "Object Detection in 20 Years: A Survey." *Proceedings of the IEEE* 111(3):257-276 (2023). DOI 10.1109/JPROC.2023.3238524 [2] R. Guedes de Souza, A.R. Panisson. "Who Thinks Best Depends on How Long You Let Them: Budget-Dependent Rankings in LLM Evaluation." arXiv:2608.12150 (2026). [3] S. Kapoor, A. Narayanan. "Leakage and the reproducibility crisis in machine-learning-based science." *Patterns* 4(9):100804 (2023). DOI 10.1016/j.patter.2023.100804 [4] B. Barz, J. Denzler. "Do We Train on Test Data? Purging CIFAR of Near-Duplicates." *Journal of Imaging* 6(6):41 (2020). [5] G.C. Cawley, N.L.C. Talbot. "On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation." *Journal of Machine Learning Research* 11(70):2079–2107 (2010). [6] A. Apicella, F. Isgrò, A. Pollastro, R. Prevete. "Don't stop me now: How Validation Criteria Affect Checkpoint Selection and Early Stopping." arXiv:2602.22107v2 (2026). [7] J. Dodge, S. Gururangan, D. Card, R. Schwartz, N.A. Smith. "Show Your Work: Improved Reporting of Experimental Results." *Proceedings of EMNLP-IJCNLP 2019*, pp. 2185–2194. <https://aclanthology.org/D19-1224/> [8] K. Gorman, S. Bedrick. "We Need to Talk about Standard Splits." *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*:2786–2791 (2019). DOI 10.18653/v1/P19-1267 [9] R. Reveles-Martínez, S. Burciaga-Sosa, J.M. Celaya-Padilla, S. Castro-Tapia, H. Luna-García, H. Morales-Magallanes, M.N. Regalado-Pérez, C. Landeros-Soriano, et al.. "Sequence-Aware Dataset Auditing for Leakage-Free Benchmarking of YOLO Detectors for Bottle Detection." *Technologies* 14(9):531 (2026). [10] W. Ruangsang, P. Pramkeaw. "Cross-Dataset Evaluation of YOLOv8 for Unmanned Aerial Vehicle Fire and Smoke Detection: Benchmark Contamination, Zero-Shot Transfer, and Onboard Deployment on a Low-Cost Airframe." *Drones* 10(8):635 (2026). [11] Y.K. Adimoolam, C. Poullis, M. Averkiou. "Data Leakage Detection and Deduplication in Large Scale Geospatial Image Datasets." *CVPR 2026 (Oral)*. https://openaccess.thecvf.com/content/CVPR2026/html/Adimoolam_Data_Leakage_Detection_and_Deduplication_in_Large_Scale_Geospatial_Image_CVPR_2026_paper.html [12] H. Suo, H. Wang, Y. Li. "Checkpoint Selection and Evaluation in EEG Emotion Recognition." arXiv:2607.27655 (2026). [13] Smith JE, Winkler RL. The optimizer's curse: skepticism and postdecision surprise in decision analysis.

Management Science **52**(3):311–322, 2006. doi:10.1287/mnsc.1050.0451. [14] M. R. R. Roktim. "When Is a Molecule a Duplicate? Identity Policy Determines What a Benchmark Audit Finds." *ChemRxiv* preprint (2026). DOI 10.26434/chemrxiv.15009099/v1 [15] S. Roth. "Which Leakage Types Matter? A Quantitative Landscape Across 2,047 Benchmark Datasets." arXiv:2604.04199 (2026). [16] G. Jocher, A. Chaurasia, J. Qiu. *Ultralytics YOLO* (v8.4.120) [computer software] (2023; pinned release 2026). <https://github.com/ultralytics/ultralytics> (accessed 2026-09-12); the same project's official course states the split-independence rule normatively: Ultralytics Academy, "Splits that Tell the Truth," <https://academy.ultralytics.com/courses/computer-vision-foundations/splits-that-tell-the-truth> (accessed 2026-09-26). [verified: software release tagged v8.4.120 as used in this study] [17] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. "Do ImageNet Classifiers Generalize to ImageNet?" *Proceedings of the 36th International Conference on Machine Learning*, PMLR **97**:5389–5400 (2019). [18] G. Zhang, K. Zhao. "Winning by Peeking: Unenforced Budgets and Test-Set Selection Inflate Short-Budget AutoML Comparisons." arXiv:2608.07303 (2026). [19] Taylor J, Tibshirani RJ. Statistical learning and selective inference. *Proceedings of the National Academy of Sciences* **112**(25):7629–7634, 2015. doi:10.1073/pnas.1507583112. [20] Benjamini Y, Yekutieli D. False discovery rate-adjusted multiple confidence intervals for selected parameters. *Journal of the American Statistical Association* **100**(469):71–81, 2005. doi:10.1198/016214504000001907. [21] M. Everingham, L. Van Gool, C.K.I. Williams, J. Winn, A. Zisserman. "The Pascal Visual Object Classes (VOC) Challenge." *International Journal of Computer Vision* **88**(2):303-338 (2009). [22] Kaggle data record, "Face mask detection" (CC0; 853 images, 3 classes, PASCAL-VOC XML): <https://www.kaggle.com/datasets/andrewmvd/face-mask-detection>. No DataCite or Mendeley Data record carries this title (sweep, 2026-10-09). [23] J. Dodge, G. Ilharco, R. Schwartz, A. Farhadi, H. Hajishirzi, N. Smith. "Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping." arXiv:2002.06305 (2020). DOI 10.48550/arXiv.2002.06305 [24] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Sepah, E. Raff, et al. "Accounting for Variance in Machine Learning Benchmarks." arXiv:2103.03098 (2021). DOI 10.48550/arXiv.2103.03098 [25] B.L. Welch. "The Generalization of 'Student's'

Problem when Several Different Population Variances are Involved." *Biometrika* 34(1/2):28 (1947). [26] K. Kann, K. Cho, S.R. Bowman. "Towards Realistic Practices In Low-Resource Natural Language Processing: The Development Set." *Proceedings of EMNLP-IJCNLP 2019*; arXiv:1909.01522 (2019). [27] R. Roelofs, V. Shankar, B. Recht, S. Fridovich-Keil, M. Hardt, J. Miller, L. Schmidt. "A Meta-Analysis of Overfitting in Machine Learning." *Advances in Neural Information Processing Systems* **32**:9179-9189 (2019). [28] Y. Benjamini, Y. Hochberg. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society Series B: Statistical Methodology* 57(1):289-300 (1995). [29] *YOLOX* [computer software]. <https://github.com/Megvii-BaseDetection/YOLOX> (pinned at commit 6ddff4824372, branch main, accessed 2026-09-16). [30] *Detectron2* [computer software]. <https://github.com/facebookresearch/detectron2> (pinned at commit a2f4a8771ab7, branch main, accessed 2026-09-16). [31] *MMDetection* [computer software]. <https://github.com/open-mmlab/mmdetection> (pinned at commit cfd5d3a985b0, branch main, accessed 2026-09-16). [32] *DETR* [computer software]. <https://github.com/facebookresearch/detr> (pinned at commit 29901c51d7fe, branch main, accessed 2026-09-16). [33] *PaddleDetection* [computer software]. <https://github.com/PaddlePaddle/PaddleDetection> (pinned at commit b25522a0f4bd, branch release/2.9, accessed 2026-09-16). [34] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, Piotr Dollár. "Microsoft COCO: Common Objects in Context." arXiv:1405.0312 (2014). DOI 10.48550/arXiv.1405.0312 [35] S. Shao, Z. Li, T. Zhang, C. Peng, G. Yu, X. Zhang, J. Li, J. Sun. "Objects365: A Large-Scale, High-Quality Dataset for Object Detection." *2019 IEEE/CVF International Conference on Computer Vision (ICCV)* 8429-8438 (2019). [36] A. Kuznetsova, H. Rom, N. Alldrin, J. Uijlings, I. Krasin, J. Pont-Tuset, S. Kamali, S. Popov, et al.. "The Open Images Dataset V4." *International Journal of Computer Vision* 128(7):1956-1981 (2020). [37] G.-S. Xia, X. Bai, J. Ding, et al. "DOTA: a large-scale dataset for object detection in aerial images." *CVPR* 2018, pp. 3974-3983. DOI 10.1109/CVPR.2018.00418. [38] P. Zhu, L. Wen, D. Du, et al. "Detection and

tracking meet drones challenge." *IEEE TPAMI* 44(11):7380-7399 (2022). DOI 10.1109/TPAMI.2021.3119563. (VisDrone.) [39] J. Wang, W. Yang, H. Guo, R. Zhang, G.S. Xia. "Tiny Object Detection in Aerial Images." *2020 25th International Conference on Pattern Recognition (ICPR)* 3791-3798 (2021). DOI 10.1109/icpr48806.2021.9413340 [40] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, et al.. "The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking." *Lecture Notes in Computer Science* 375-391 (2018). DOI 10.1007/978-3-030-01249-6_23 [41] Darius Lam, Richard Kuzma, Kevin McGee, Samuel Dooley, Michael Laielli, Matthew Klaric, Yaroslav Bulatov, Brendan McCord. "xView: Objects in Context in Overhead Imagery." arXiv:1802.07856 (2018). DOI 10.48550/arXiv.1802.07856 [42] Ke Li, Gang Wan, Gong Cheng, Liqu Meng, Junwei Han. "Object Detection in Optical Remote Sensing Images: A Survey and A New Benchmark." arXiv:1909.00133 (2019). DOI 10.48550/arXiv.1909.00133 (DIOR benchmark) [43] G. Cheng, J. Han, P. Zhou, L. Guo. "Multi-class geospatial object detection and geographic image classification based on collection of part detectors." *ISPRS Journal of Photogrammetry and Remote Sensing* 98:119-132 (2014). [44] Xu et al. "Detection of safety helmet wearing based on improved SSD." *Journal of Physics: Conference Series* (2019). Dataset: <https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset> [45] Fusheng Yu, Jiang Li, Xiaoping Wang, Shaojin Wu, Junjie Zhang, Zhigang Zeng. "Large, Complex, and Realistic Safety Clothing and Helmet Detection: Dataset and Method." arXiv:2306.02098 (2023); journal version in *IEEE Transactions on Instrumentation and Measurement* (2024). DOI 10.48550/arXiv.2306.02098 (SFCHD / SFCHD-SCALE) [46] S. Ge, J. Li, et al. "Detecting masked faces in the wild with LLE-CNNs." *CVPR* 2017, pp. 426-434. DOI 10.1109/CVPR.2017.53. (Cited only to delimit the MAFA packaging difference.) [47] S. Yang, P. Luo, C.C. Loy, X. Tang. "WIDER FACE: A Face Detection Benchmark." *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 5525-5533 (2016). [48] Shuai Shao, Zijian Zhao, Boxun Li, Tete Xiao, Gang Yu, Xiangyu Zhang, Jian Sun. "CrowdHuman: A Benchmark for Detecting Human in a Crowd." arXiv:1805.00123 (2018). DOI 10.48550/arXiv.1805.00123 [49] P.V.A.B. de Venâncio, A.C. Lisboa, A.V. Barbosa. "An automatic fire detection system based on deep convolutional

neural networks for low-power, resource-constrained devices." *Neural Computing and Applications* 34(18):15349-15368 (2022). DOI 10.1007/s00521-022-07467-z [50] L. Cascone, M. Nappi, C. Pero, X. Wang. "A framework for bias-aware dataset evaluation in soft facial attribute recognition." *Pattern Recognition* 172:112416 (2026). DOI 10.1016/j.patcog.2025.112416 [51] Y. Huang, Z. Zhang, Y. Huang, Q. Wu, H. Huang, Y. Zhong, L. Wang. "Customized meta-dataset for automatic classifier accuracy evaluation." *Pattern Recognition* 146:110026 (2024). DOI 10.1016/j.patcog.2023.110026 [52] J. Sweeney. "Optimizer Memory Makes Shuffle Order a First-Order Source of Fine-Tuning Noise." arXiv:2606.29554 (2026). [53] E. Pintelas, I.E. Livieris. "GeNeX: Genetic Network eXperts framework for addressing Validation Overfitting." *IEEE Transactions on Neural Networks and Learning Systems* (2026); arXiv:2603.11056. DOI 10.48550/arXiv.2603.11056 [54] Y. Tian, Q. Ye, D. Doermann. "YOLOv12: attention-centric real-time object detectors." arXiv:2502.12524 (2025). [55] N. Ajroldi, A. Orvieto, J. Geiping. "When, Where and Why to Average Weights?" arXiv:2502.06761 (2025).